

$$E2: 2^{2^n}, 2^{n^2}, n^2 \log n, n \cdot n^{2^n}$$

¿Cual crece mas rapido? $2n^2$ & n^{2^n}

Usando limites

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)}; f(n) = o(g(n)) = 2n^2 = o(n^{2^n})$$

$$\lim_{n \rightarrow \infty} \frac{2n^2}{n^{2^n}}$$

$$n^{2^n} = (n^2)^n = \lim_{n \rightarrow \infty} \frac{2n^2}{(n^2)^n}$$

$$\frac{2n^2}{n^{2^n}} = \frac{2n^2}{(n^2)^n}$$

$$\frac{2n^2}{(n^2)^n} = \frac{2}{(n^2)^{n-1}}$$

$$\frac{2n^2 \text{ (polynomial)}}{(n^2)^n \text{ (exponencial en } n\text{)}}$$

$$\lim_{n \rightarrow \infty} \frac{2n^2}{n^{2^n}} = \emptyset$$

n^{2^n} crece mucho mas que $2n^2$

Cuando $n \rightarrow \infty$

$$(n^2)^{n-1} \rightarrow \infty$$

$$2n^2 = o(n^{2^n})$$

$$\lim_{n \rightarrow \infty} \frac{2}{(n^2)^{n-1}} = 0$$

n	$2n^2$	n^{2^n}
2	8	16
3	18	729
4	32	65,536
5	50	9,765,625

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